

SENIOR HCI PROJECT · 2025–2026

Wasla. Your Smart Subscription Hub.

Your Smart Subscription Hub — track, manage, and save



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Umm Al-Qura University



Flutter · Firebase



WCAG 2.1 AA / AAA

SUPERVISOR · المشرف

Dr. Reem Saleh Alshaikh · د. ريم الشيخ

One hub. Every subscription. Zero surprise charges.

Wasla is a smart bilingual subscription manager for Saudi Arabia and the GCC — designed to fix the fragmented way people track recurring payments across streaming, software, gym, education, and family services.



Problem

Subscriptions are scattered across emails, apps, and bank statements. Users lose track and pay for what they don't use.



Goal

Centralise every recurring payment, surface the real monthly / yearly cost, and alert users before each renewal.



Solution

A Flutter + Firebase mobile app with native RTL Arabic, deterministic AI chatbot, and guided cancellation.



Outcome

Validated with 221 survey participants and 5 think-aloud users. WCAG AAA contrast. 6–8 weeks to ship.

Subscriptions are bleeding Saudi wallets — silently.

Validated with 221 survey participants across Saudi Arabia. 50.7% experienced at least one unexpected charge.

96.9%

Want renewal reminders

لم يكن لديهم تطبيق يقدم تذكيرات قبل التجديد.

50.7%

Hit by surprise charges

تعرضوا لرسوم غير متوقعة على الأقل مرة واحدة.

76%

Forget active subscriptions

ينسون اشتراكات نشطة ويدفعون بدون استخدام.

84%

Want Arabic first

RTL. يطلبون تطبيقاً عربياً أصيلاً لمواجهة

A 4-pillar ecosystem, designed for the GCC.

One hub. Four orbiters. End-to-end control over every subscription in your life — digital, physical, family, or business.



TRACK • كل اشتراك في مكان

Live renewal calendar + customisable reminders per subscription.



ANALYSE • لوحة إنفاق ذكية

Monthly / yearly spending toggle with category breakdown.



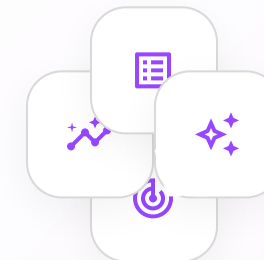
ASSIST • مساعد عربي فوري

Deterministic Arabic chatbot with curated FAQ responses.



CANCEL • إلغاء ميسر

Step-by-step cancellation guide for any service provider.



Eight capabilities, one super-app.

Each feature is backed by survey data, think-aloud tests, and a clean design system — no fluff, no scope creep.



Smart Add

إضافة ذكية • auto-categorise + price intelligence



Analytics

تحليلات • monthly / yearly spending toggle



Renewal Alerts

تنبيهات • 3-day default, customisable per sub



Smart Offers

عروض ذكية • personalised deals on top cats



Cancel Guide

دليل الإلغاء • step-by-step in-app



AI Assistant

مساعد ذكي • deterministic Arabic chatbot



Favourites

المفضلة • pin important subs for quick access



Multi-Channel

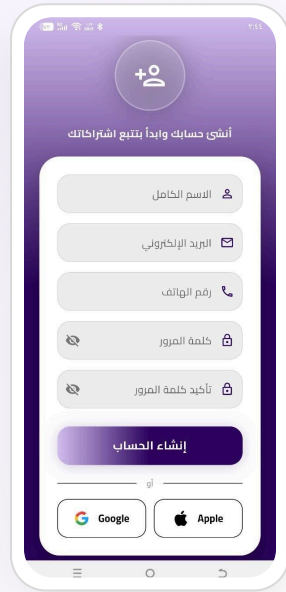
دعم متعدد • WhatsApp • Email • Phone

Eight high-fidelity screens, validated in Arabic.

Built in Flutter · Material 3 · with native RTL support. Dashboard colors match the design system shipped in the actual app.



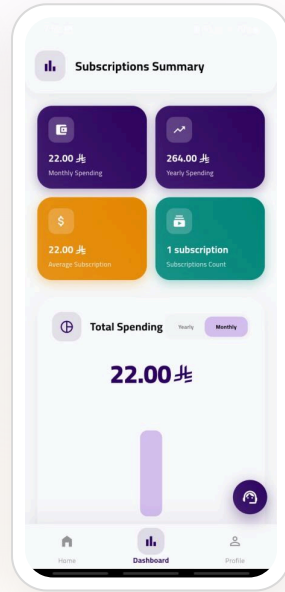
دخول · Login



تسجيل · Signup



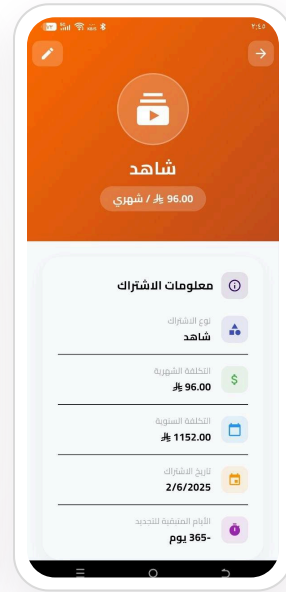
الرئيسية · Home



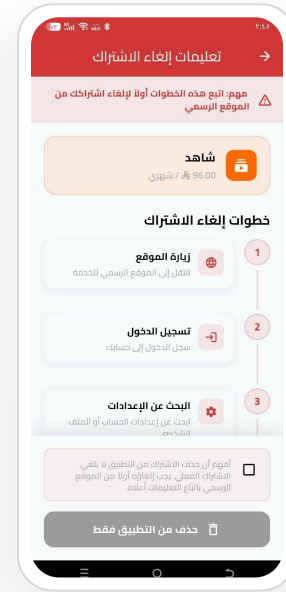
الإنفاق · Dashboard



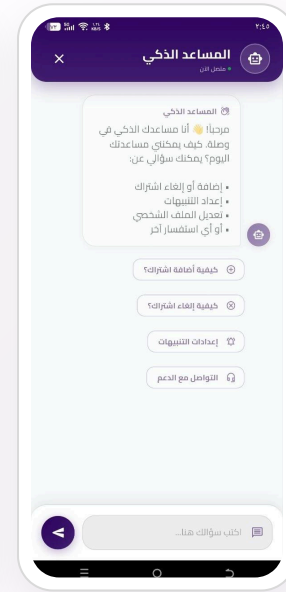
إضافة · Add



التفاصيل · Details



إلغاء · Cancel



المساعد · Chatbot

Clean Dart + Firebase. **Three actors**, one model.

The Subscription model is the heart. The Dashboard widget reads it. The Chatbot service answers questions about it. Real code from `lib/models/subscription_model.dart` on the left.

lib/models/subscription_model.dart

```
// Real Subscription model from waslaa app
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';

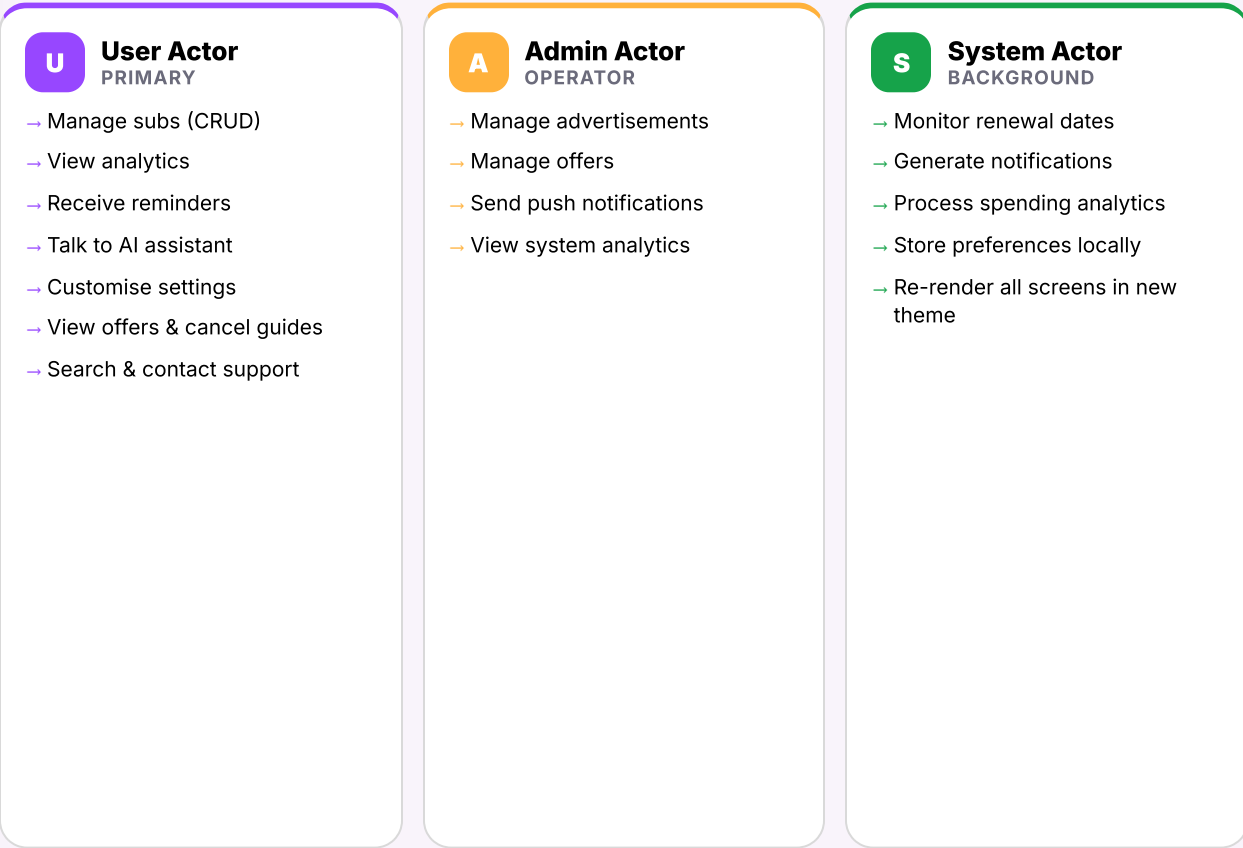
class Subscription {
  final String id;
  final String userId;
  final String name;
  final String? customName;
  final double price;
  final DateTime renewalDate;
  final DateTime? expiryDate;
  final String iconColor;
  final String? notes;
  final String? reminder;
  final String? contactInfo;

  late Color _cachedColor;

  Subscription({
    required this.id,
    required this.userId,
    required this.name,
    this.customName,
    required this.price,
```

1 Three actors, one model

The Subscription model is the single source of truth. Each actor has a focused responsibility around it.



Mixed-methods research, two GPs, 226 participants.

A multi-stage user-centred design process that informs every design decision in Wasla — from problem framing to final validation.

- 1

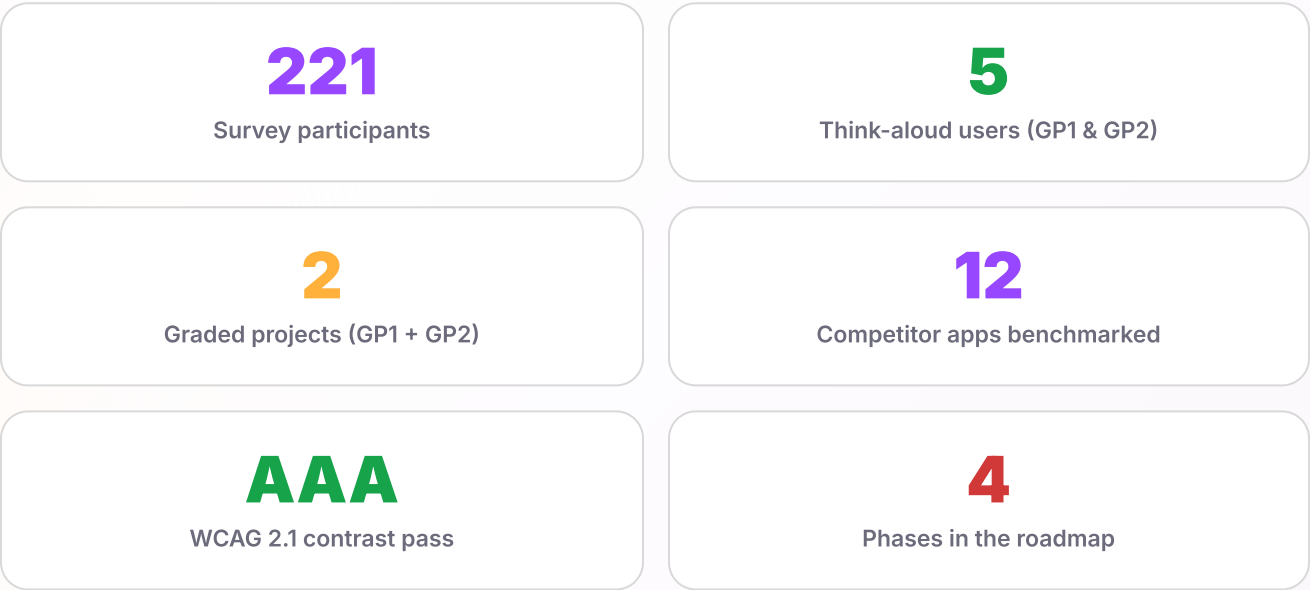
Literature & Market Review
Analysed 12 popular subscription apps to identify feature gaps in the GCC market.
- 2

Quantitative Survey · 221 participants
Online questionnaire targeting young adults (18–35) in Saudi Arabia — covered reminders, RTL, cancellation behaviour.
- 3

Qualitative Interviews · 5 users
Semi-structured interviews to deepen findings from the survey and uncover pain points around renewal tracking.
- 4

Think-Aloud Usability Test · 5 users
Conducted during GP1 and GP2 to validate prototypes and the final system, capturing confusion in real time.
- 5

Accessibility & Visual Audit
WCAG 2.1 AA / AAA contrast verified at 19.41:1, 15.42:1, 12.57:1, 11.56:1 across all primary tokens.



Think-Aloud usability testing, twice.

A formative usability test with 5 participants during both GP1 and GP2. Participants performed real tasks while verbalising thoughts, confusion, and feedback.

1

Add a new subscription

From launch, through category selection, to saving.

2

Edit an existing subscription

Modify reminder settings, price, or notes.

3

Delete a subscription

Including the confirmation flow and feedback.

4

Manually add a custom subscription

Including contact info and notes fields.

5

Display the control panel

Open the dashboard and inspect monthly / yearly view.

Test Setup

- ✓ Test name: Usability test for Wasla application
- ✓ Test type: Usability Testing, Think-Aloud Protocol
- ✓ Sample size: 5 users per round (GP1 + GP2)
- ✓ Environment: Mobile (Android), touch-based
- ✓ Prerequisites: Logged in, internet, ≥1 existing sub, notifications enabled
- ✓ Data captured: time-on-task, error count, verbal protocol, post-task survey
- ✓ Iteration cycle: find issues → fix in GP2 → re-test → ship

GP1 → GP2: measurable improvement on every aspect.

A side-by-side comparison of GP1 (prototype) versus GP2 (final system) across the same six usability aspects.

ASPECT	GP1 PROTOTYPE	GP2 SYSTEM	IMPROVEMENT	STATUS
Add Subscription	Add button small & unclear, up to 55s to complete.	Most users added subs successfully.	Improved button visibility and easier navigation.	Pass
User Interface	Small fonts, unclear layout, unnecessary dashboard elements.	Dashboard appreciated — clear and organized.	Better UI organization and dashboard presentation.	Pass
Reminder Editing	Feature not available in GP1 testing.	Most users successfully edited reminders.	Added reminder editing with acceptable usability.	Pass
Delete / Cancel	Process simple, but no direct unsubscribe links.	Most users deleted subs; some felt confirmations were excessive.	Improved deletion flow and user guidance.	Pass
Subscription Info	Some users confused about expiry & renewal tracking.	Dashboard clearly shows monthly & yearly expenses.	Improved clarity of subscription tracking.	Pass
Custom Subscription	Feature not available in GP1 prototype.	Users could manually add custom subscriptions.	More flexible subscription management.	Pass



All 6 aspects passed

Zero blocking issues, no critical errors.



Minor follow-ups

Fewer confirmations, clearer custom-sub labels.



Visual polish

One user requested a colour & UI refresh — shipped in GP2.

Wasla closes every meaningful gap.

Benchmarked against the three most popular subscription managers in global app stores.

BillBot • Subby • Bobby Most popular subscription apps	
Arabic language support	X
Smart personalised offers	X
Cancellation guidance	PARTIAL
Spending analytics	PARTIAL
Personalized settings	PARTIAL
AI chatbot assistant	X
Dark / Light mode	X
Font size customisation	X
Multi-channel support	X
Native RTL interface	X

Wasla • وصلة Built for Saudi Arabia & the GCC	
Arabic language support	✓
Smart personalised offers	✓
Cancellation guidance	✓
Spending analytics	✓
Personalized settings	✓
AI chatbot assistant	✓
Dark / Light mode	✓
Font size customisation	✓
Multi-channel support	✓
Native RTL interface	✓

From sign-up to saved money, in 5 steps.

A linear, predictable flow that 5 think-aloud users completed without guidance during GP2 testing.



Register / Login

Email + password, Google or Apple sign-in.



Add Subscription

Category, price, dates, reminder settings.



View Dashboard

Monthly / yearly spending with toggle.



Get Reminders

Notified 3 days before each renewal.



Manage & Save

Edit, cancel, smart offers, chatbot.

A modern, accessible foundation.

Cross-platform Flutter client, Firebase backend, deterministic AI, Material Design 3 with custom Wasla tokens.



Flutter / Dart

iOS + Android from one codebase



Material 3

Custom Wasla design tokens



Firebase

Auth · Firestore · Functions



FCM

Renewal alerts and pushes



Deterministic AI

Curated Arabic responses



Figma

High-fidelity prototypes



Provider / BLoC

State management



Cloud Functions

Reminder scheduling

From prototype to a smarter future.

GP1 and GP2 are validated and shipped. Phase 3 unlocks the platform for personal finance.

GP1 • COMPLETED

Core CRUD + Dashboard

Add / edit / delete subscriptions, basic dashboard, reminder, login flow. Validated with 5 users.

GP2 • COMPLETED

AI + Offers + Cancellation

Deterministic Arabic chatbot, smart offers, step-by-step cancel, reminder editing, improved UI.

PHASE 3 • FUTURE

Bank Integration

Connect bank and payment platforms, auto-discover subscriptions, detect silent renewals.

PHASE 4 • FUTURE

Advanced AI + Marketplace

Spending predictions, marketplace, family sharing, smarter recommendations.

Designed around **real users, real data.**

A research-driven HCI project — every design decision is backed by a survey, an interview, or a usability test.

221

Survey Participants

Young adults & professionals across Saudi Arabia.

96.9%

Want Reminders

And didn't have them in any existing app.

50.7%

Hit by Surprise

Experienced at least one unexpected charge.

19.41:1

WCAG Pass

Contrast ratio verified AA & AAA.

14 • TEAM • الفريق

The people behind Wasla.

Six future HCI specialists from Umm Al-Qura University, guided by Dr. Reem Saleh Alshaikh at the College of Computing. Every name, every role, every credit.

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Full technical documentation, **publicly available.**

A complete technical reference lives at `/docs` — design system, file tree, code samples, deployment guide, and accessibility audit.



Design System

12 brand tokens, light & dark variants, typography, spacing, motion, elevation.



File Tree

lib/models, lib/services, lib/providers, lib/screens, assets, with annotated responsibilities.



Code Samples

Subscription model, SpendingChart, ChatbotService intent matcher, AuthService.



Architecture

Three actors, Firebase integration, FCM reminders, Cloud Functions for scheduling.



Accessibility

WCAG 2.1 AAA contrast ratios, font scaling, dark mode, RTL audit, screen reader labels.



Deployment

Vercel-ready static site, GitHub Actions CI, security headers, clean URLs, asset caching.

END • النهاية

Thank You.

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🔗 Flutter • Firebase • Dart



Umm Al-Qura University • College of Computing

Presented by the Wasla team • 2025–2026